Combustion Dynamics of Turbulent Swirling Flames

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In the present paper the influence of a periodic excitation of the mass flow rate, varied in frequency and in amplitude, on the isothermal flow field and on the flame characteristics of swirl burners with different swirl intensities $S_{0,th}$ is investigated. At first the fluid dynamical conditions for the formation of ring-vortex structures in the burner near flow field are determined. The results indicate that the minimum level of excitation of the mass flow rate for vortex formation decreases hyperbolically with increasing frequency of pulsation and characteristic Strouhal number, respectively. In further studies, the dynamical behavior of lean-premixed swirl flames is investigated, whereas the fuel gas/air mixture mass flow rate at the burner exit was modulated sinusoidally. The dynamical behavior of the investigated flames is found to be dominated by two different effects in certain frequency ranges: For moderate pulsation frequencies ($f_{puls} \lesssim 50$ Hz) the detected periodic heat release rate of pulsated, premixed swirl flames is dominated by an effect that inhibits the strong entrainment of ambient medium in comparison with the corresponding quasi steady-state swirl flames. With increasing frequency ($f_{puls} \gtrsim 50$ Hz) this effect will be overlaid by the periodical formation of ring-vortices entraining additional ambient medium and interfering with the combustion process. The physical understanding of the frequency-dependent flame dynamics (flame transfer function) on periodic disturbances is indispensable for the prediction of the formation of combustion-driven oscillations that occur in technical combustion systems (e.g., gas turbines, industrial combustors). It is also evident for the development of methods to prevent or suppress periodic combustion instabilities. The results of the flame transfer measurements presented in this paper will lead to a basic understanding of the formation and reaction of large-scale coherent vortex structures in turbulent flames, that are well known as drivers of combustion instabilities. © 2002 by The Combustion Institute

NOMENCLATURE

d	diameter
f	frequency
$\lvert F \rvert$	amplitude response of the flame transfer function
$\dot{m}$	mass flow rate
p	pressure
Pu	pulsation level, degree of excitation
$\dot{Q}_{th}$	thermal load of the flame
Re	Reynolds number
Str	Strouhal number
S	swirl intensity
t	time
u	velocity
U	photomultiplier voltage as a measure of OH*-radiation intensity
$\dot{V}$	volume flow rate
x	axial position

Greek Symbols

δ	vane angle of the swirl generator
Δ	interval
φ	phase angle function of the flame transfer function
λ	$(=1/\phi)$ air equivalence ratio
λ	wavelength
ϕ	(fuel-air) equivalence ratio
ω	$(=2 \cdot \pi \cdot f_{puls})$ angular frequency

Subscript

0,th	theoretical value at the burner exit
b	at the burner exit
cc	combustion chamber
d	delay
dis	disturbance
eff	effective, in the main flow direction
eq	equivalent
fl	flame
mix	fuel gas/air mixture
n	at normal conditions (T_n = 273 K, p_n = 1013 mbar)
OH,max	maximum of the OH*-radiation intensity
p	period of pulsation
pm	photomultiplier

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premix	the fuel gas/air mixture at the burner exit
puls	pulsation
ref	reference value
rms	rms-value, containing periodical and turbulent contributions
tr	trigger
x	in axial direction

Superscripts

$\bar{}$	mean value
$\sim$	periodic fluctuations
$\hat{}$	amplitude
*	excited state

INTRODUCTION

The development of combustion systems to minimize the flue gas emissions (NO_x, CO, HC, etc.) and to increase the system's efficiency (e.g., gas turbine) is combined with technical modifications or new combustor design. This also includes modifications of the combustion process using premixed systems and highly turbulent swirl stabilized flames. The application of these modifications makes high-intensity combustion and therewith, combustion systems with higher reaction densities possible. Increasing stability problems caused by the appearance of combustion-driven oscillations are observed to be linked with these improvements. The induced pressure oscillations can lead to the total damage of the installations and therefore, the manufacturers are forced to operate these combustors under not acceptable conditions. The best way to develop measures to suppress or to eliminate these combustion driven oscillations is, in contrast to time- and cost-intensive trial and error solutions, to realize and to understand the relevant feedback mechanism, that is necessary to sustain the periodic pressure oscillations in the combustion chamber.

Possible feedback mechanisms for the sustainment of combustion driven oscillations in highly turbulent, premixed combustion systems are schematically illustrated in Fig. 1. As a very important mechanism driving these combustion instabilities the in-phase formation and combustion of large-scale coherent vortical structures were identified [1–5]. Another important mechanism considers the role of equivalence ratio fluctuations caused by the pressure oscillations coupling with the mixing unit in industrial combustion systems. For a detailed description of that mechanism we refer to literature [6–8]. This mechanism doesn't play a role in the current experiments, as the equivalence ratio was kept constant, ensuring a homogeneous, perfect mixture during the flame experiments.

The formation of ring-vortex structures is a fundamental fluid dynamical phenomenon, that can always be observed together with the occurrence of technical relevant, not tolerable pressure oscillations in highly turbulent combustors. The appearance of these ring-vortices was found for non-premixed as well as for premixed combustion systems and is also independent of the applied flame type (jet-, bluff body-, or swirl stabilized flames). At a particular combination of amplitude and frequency of the pressure fluctuations ($\hat{p}_{dis}$, f_{dis}) these toroidal vortices are formed, rolling-up the outer regions of the fuel gas/air mixture flow together with additional entrained ambient medium [2, 9–11]. Therefore, the mixing and reaction fields of the steady-state flame are changing totally. The reaction of the fuel in the vortex causes a fluctuation of the heat release rate of the flame. The resulting fluctuation of the flue gas production rate leads to a fluctuation of the static pressure in the combustion chamber (Fig. 1). In the case that the pressure fluctuations $\tilde{p}_{cc}(t)$ in the combustion chamber are in-phase with the fluctuations of the heat release rate of the flame, fulfilling Rayleigh's criterion [12], and are large enough to cover friction losses of the combustor, the fluctuation of mass flow rate at the burner exit $\tilde{\dot{m}}_{mix}$ can be amplified. This amplification of the flow rate oscillation can cause a next larger ring-vortex generation, containing a larger share of the fuel gas/air mixture (feedback loop), which in turn amplifies the fluctuation of the heat release rate and of the static pressure in the combustion chamber. This feedback loop clearly explains a mechanism for the amplification of initial small disturbances to high-amplitude pressure oscillations at proper timing conditions.

Hence, the understanding of the interaction of the periodic flow field, forced by the pressure fluctuations in the combustion chamber, and the

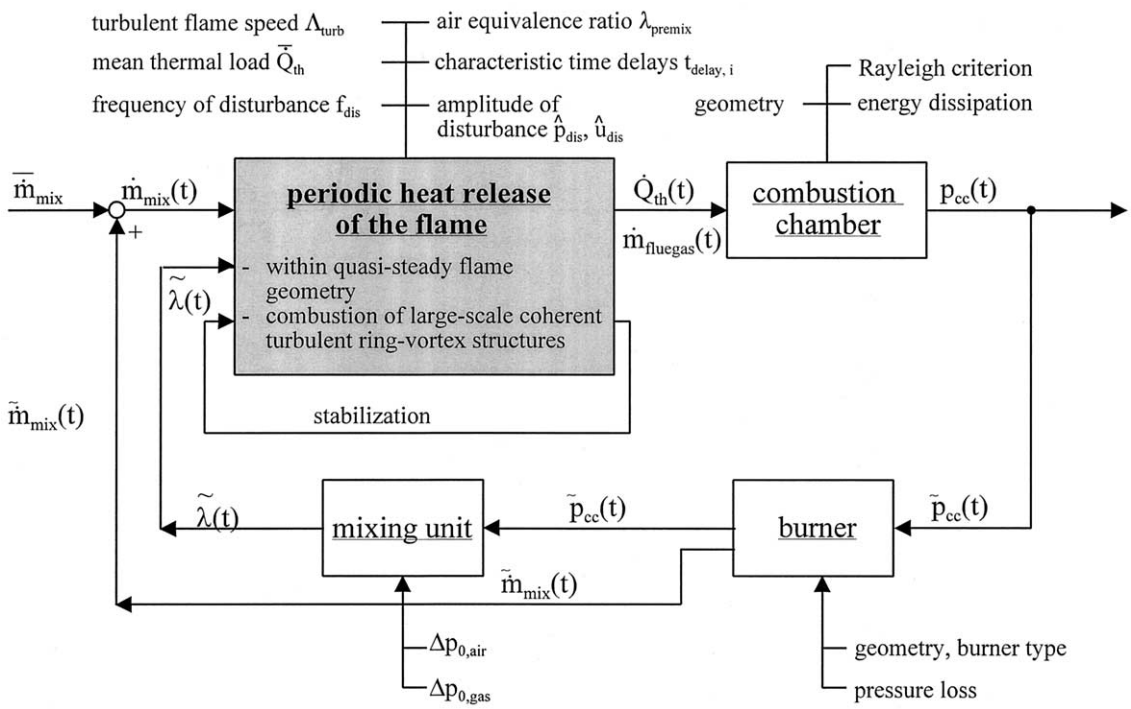

Fig. 1. Feedback mechanisms in technical combustion systems.

flame response is one of the keys to find methods to control self-excited combustion instabilities sustained by this mechanism. Investigations to determine the transfer functions of flames are based on two important aspects, that have direct influence on the appearance of periodic combustion instabilities in a burner-flame-combustion chamber assembly (Fig. 1):

1. The flame provides the energy for the amplification of initial small disturbances in the combustion system and for the sustainment of the pressure fluctuations, while it covers the friction losses of the flue gas oscillation in the combustion chamber.
2. The effective time delay between the disturbance of the mass flow through the burner nozzle and the fluctuation of the resulting heat release rate of the flame varies in dependence on numerous operation parameters of the flame (e.g., thermal power, air equivalence ratio, air-preheating). Whereas, in a defined burner and combustion chamber geometry, the time delay between heat release rate and pressure fluctuation and between pressure and mass flow modulation are more or less fixed at a given geometry of the components. Therefore, the flame is the most variable component with regard to its flexibility in closing the feedback loop.

In the present paper the formation of the large-scale toroidal vortices in isothermal swirl flows is determined in dependence of frequency and amplitude of a harmonically modulated mass flow rate at the nozzle exit. The influence of the vortical structures on the combustion process is discussed by means of measuring flame transfer functions as it was done in previous papers [4, 5]. In analogy to control theory [13] the flame transfer function describes the dynamic response characteristics of the flame in form of an amplitude response $|F_{fl}(f_{puls})|$ and a phase angle function $\varphi_{fl}(f_{puls})$. Therefore, the time functions of the flame excitation and of the resulting flame response have to be determined. In this paper the time signals correlated with the mixture mass flow rate at the burner exit (flame excitation) and with the heat release rate of the entire flame (flame response) were recorded simultaneously and analyzed to calculate the flame transfer function.

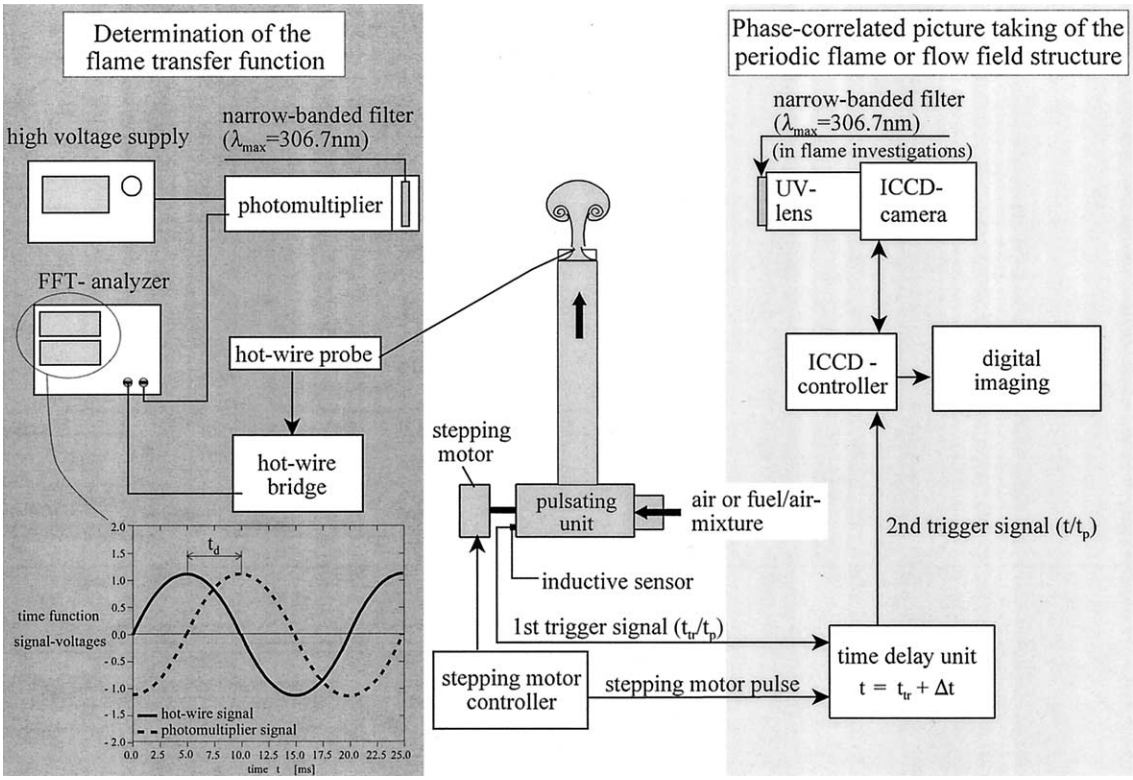

Fig. 2. Scheme of the experimental set-up and the applied measuring techniques.

The amplitude response $|F_{fl}(f_{puls})|$ is given by the ratio of the rms-value of the total heat release rate and the rms-value of the mixture mass flow rate at the burner exit at the adjusted frequency of pulsation f_{puls}, normalized by their respective rms-values of the corresponding quasi-steady flame (here: at $f_{puls} = f_{ref} = 5$ Hz). The phase angle function $\varphi_{Fl}(f_{puls})$ is related to the overall time delay $\bar{t}_{d,fl}(f_{puls})$ between the periodic fluctuations of the fuel gas/air mixture flow at the nozzle exit and the resulting periodic fluctuation of the reaction rate of the entire turbulent flame.

EXPERIMENTAL SETUP AND MEASUREMENT TECHNIQUES

The investigations of pulsated isothermal flow fields and natural gas-fired, premixed swirl flames were carried out by using swirl burners with different swirl intensities $S_{0,th} = 0.21$, 0.45 and 0.79. The swirl intensities $S_{0,th}$ were calculated from the geometric data of the axial vane swirlers ($d_b = 57$ mm, $d_{bluff\text{-}body} = 35$ mm) with different vane angles ($\delta_{vane} = 15$, 30, and 45 degree). The calculation equation for $S_{0,th}$ is given in [14]. Figure 2 shows schematically the experimental setup and the applied measurement techniques.

An adjustable part of the total fuel gas/air mixture mass flow was modulated nearly sinusoidally in time using a pulsating unit. This pulsating unit can generate periodic oscillations of the mass flow rate at the burner nozzle exit within a frequency range between 1 to 350 Hz with continuously adjustable amplitudes up to 70% of the mean mass flow rate. A detailed description of this pulsating unit is given in previous papers [4–6, 10]. The pulsation frequency can be chosen by adjusting the rotation of the stepping motor with an accuracy of 0.01 Hz. In the case of flame investigations a decoupling chamber was positioned between the pulsating unit and the mixing section of the reactants. This decoupling chamber prevents the periodic pressure oscillations, generated by the

pulsating unit, from influencing the mass flow adjustment with the flow meters and the fuel gas/air mixing process [6]. Therewith, the equivalence ratio of the fuel gas/air mixture mass flow through the burner was kept constant during all flame experiments.

The applied measurement techniques were:

- Hot-wire anemometry to determine the periodic mass flow fluctuation at the burner exit (flame excitation).
- A photomultiplier detecting the overall radiation intensity of the flame (flame response) at a wavelength $\lambda = 306.7$ nm, emitted by excited OH*-radicals from the reaction zone of the flame. This wavelength corresponds to the (0, 0) band of the $2\Sigma - 2\Pi$ electronic state transition of the diatomic OH molecule [15].
- Phase-correlated picture-taking of the periodically changing flow field or flame structures. Using a trigger signal, generated by an inductive sensor, which is mounted in the pulsating unit, phase-locked picture-taking at a fixed relative time t_{tr}/t_p within the period of excitation is possible. An additional time delay unit allows to observe the time history of the periodically disturbed flow field and flame structure at different moments t/t_p within the period of excitation. To realize short exposure times ($\Delta t_{exp} = 10$–$100\ \mu$s), that are necessary for watching the flow field or the flame structure at particular moments, an intensified CCD-Camera (512 × 512 Pixel up to 16 bit A/D at 1 MHz) was used (Fig. 2).

Determination of Fluid Dynamical Conditions for Vortex Formation

The determination of the fluid dynamical conditions to generate first-time coherent vortex structures in a periodically modulated isothermal flow field was carried out by using an in situ calibrated single tungsten hot-wire sensor (2 mm in length and 5 μm in diameter) located between two vanes of the swirler at the burner exit. With the help of the measured time-dependent hot-wire signal voltage $U_{hw}(t)$ the mean value $\bar{\dot{m}}_b$ and the rms-value $\dot{m}_{b,rms}$ of the mass flow rate $\dot{m}_b(t)$ at the burner exit can be calculated. It has to be mentioned that this calculation is based on some assumptions: constant velocity profile in radial direction at any moment within the period of excitation, rotational symmetry, incompressible flow. Additionally the rms-value includes to a small extent also turbulent fluctuations. The level of excitation Pu, which is together with the pulsating frequency f_{puls} the dominating parameter to describe the formation of coherent vortex structures in periodically disturbed flow fields, can be calculated using Eq. 1 [4–6]:

$$\mathrm{Pu} = \frac{\dot{m}_{b,rms}}{\bar{\dot{m}}_b} \tag{1}$$

At the beginning of the investigations the mean mass flow rate $\bar{\dot{m}}_b$ and the frequency f_{puls} were adjusted. The rectangular part of the exit area of the pulsating unit (bypass flow) [4–6, 10] was opened completely, whereas the amplitude of the periodic excitation of the mixture mass flow was set to minimum. After that, the pulsation level Pu was increased continuously by closing the bypass area of the pulsating unit while the frequency of pulsation f_{puls} was kept constant. At the same time the flow field of the burner near region (~120–150 mm), visualized by the means of a tracer medium and laser light-sheet technique, has been observed. The critical pulsation level Pu_{crit} was reached in the case that a first-time coherent vortex ring structure was formed at any moment within the period of disturbance. In this way the critical pulsation level Pu_{crit}, characterizing the appearance of these toroidal structures, was determined for different pulsation frequencies f_{puls} at burners with different swirl intensities and at different mean flow rates $\bar{\dot{m}}_b$ [16].

Determination of the Flame Transfer Function

For the measurements of flame transfer functions (Fig. 2) the determination of the mean- and rms-value of the pulsated flow rate (flame excitation) was conducted again by hot-wire anemometry. The position of the hot-wire was located 15 mm upstream of the burner exit between two vanes of the swirler. This was necessary to protect the hot-wire sensor from destruction caused by a flame flashback during the pulsated flame investigations. For the cap-

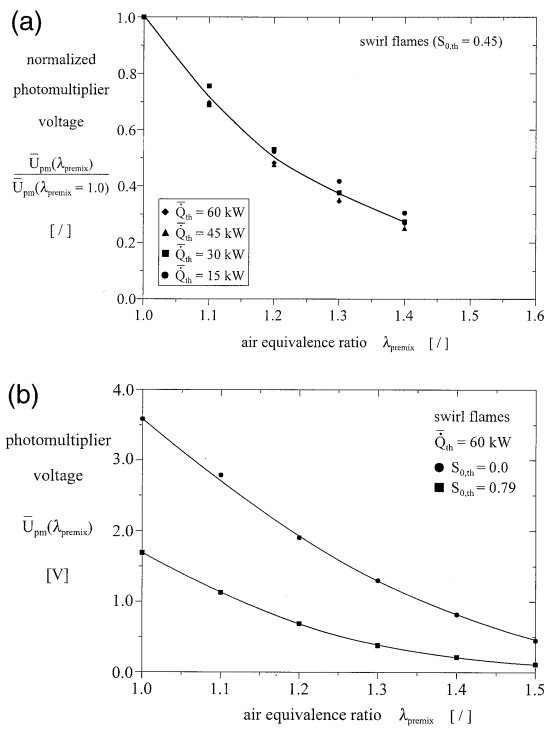

Fig. 3. Normalized OH*-radiation intensities of swirl flames with different thermal load (a). OH*-radiation intensities of swirl flames with different swirl intensities (b).

ture of the flame response a photomultiplier detecting the overall radiation of the excited OH*-radicals at 306.7 nm was used. Because of the strong dependence of this radiation intensity U_{pm} on the reaction rate $\sim\dot{Q}_{th}$ and on the air equivalence ratio λ_{premix} $(=1/\phi)$, because of their influence on flame geometry and flame temperature, respectively, their correlation was calibrated using steady-state non-enclosed premixed swirl flames.

In Fig. 3a the non-linear dependence of the radiation intensity on the air equivalence ratio λ_{premix} is shown. The individual data were normalized with the corresponding maximum value of radiation intensity at stoichiometric mixture of each thermal load level $\bar{\dot{Q}}_{th}$. The correlation between the heat release rate and the air equivalence ratio for different swirl intensities is shown in Fig. 3b. The radiation intensity is obviously lower for increasing swirl intensities. This is caused by the strongly intensified entrainment of ambient air with increasing swirl [14]. Therefore, the OH*-radiation intensities measured for the swirl intensity $S_{0,th} = 0.0$ (vane angle of the swirl generator equals zero) is the maximum level of OH*-radiation of steady-state "swirl" flames with corresponding thermal load and air equivalence ratio values. These correlations are used later on for the physical interpretation of the frequency-dependent reaction behavior of pulsated swirl flames.

The definition of the amplitude response of the flame $|F_{fl}(f_{puls})|$ for periodical excitation is given in Eq. 2 [4, 5]:

$$|F_{fl}(f_{puls})| = 20 \cdot \log \frac{\dfrac{U_{pm,rms}(f_{puls})}{U_{pm,rms}(f_{ref})}}{\dfrac{u_{b,rms}(f_{puls})}{u_{b,rms}(f_{ref})}} \tag{2}$$

The frequency-dependent rms-values of the simultaneously measured time functions of the mass flow $u_{b,rms}(f_{puls})$ and the UV-radiation $U_{pm,rms}(f_{puls})$ are normalized with their reference values $u_{b,rms}(f_{ref})$ and $U_{pm,rms}(f_{ref})$, taken at a frequency f_{ref}. At this reference frequency (f_{ref} = 5 Hz) the slowly pulsated flame behaves in every moment within the period of excitation as a steady-state swirl flame with the corresponding mean thermal load and air equivalence ratio with regard to the mean OH*-radiation intensity of the entire flame.

The phase angle function $\varphi_{fl}(f_{puls})$ between the two periodical signals was calculated using Eq. 3 [4, 5]:

$$\varphi_{fl}(f_{puls}) = -\bar{t}_{d,fl}(f_{Puls}) \cdot f_{Puls} \cdot 360° \tag{3}$$

The frequency-dependent time delay $\bar{t}_{d,fl}(f_{puls})$ was determined directly by measuring the cross correlation function of the hot-wire bridge voltage and the photomultiplier signal (see Fig. 2).

EXPERIMENTAL RESULTS AND DISCUSSION

Investigations of the Fluid Dynamical Conditions for Ring-Vortex Formation in Pulsated Isothermal Swirl Flows

The results found for periodically excited isothermal flow fields of swirl burners with different swirl intensity and various mass flow rate are shown in Fig. 4. The pulsating level Pu_{crit}, that is necessary to generate periodically toroidal

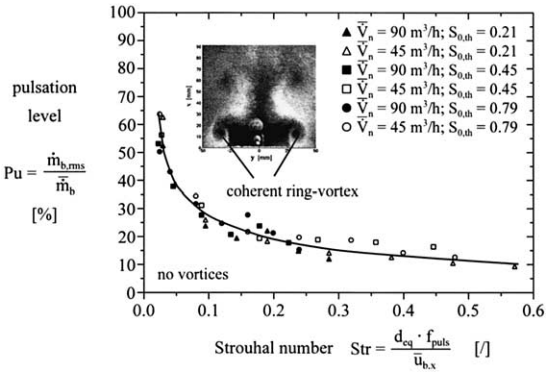

Fig. 4. Dynamical flow conditions (Pu, Str) characterizing the formation of large-scale coherent vortices in periodically pulsated swirl flows.

structures at a given pulsation frequency f_{puls}, decreases hyperbolically for all investigated flows with increasing frequency f_{puls}. The changes of the parameter combination (Pu, $f_{puls})_{crit}$ for different swirl intensities ($S_{0,th}$) and mass flow rates ($\bar{m} \sim \bar{V}_N$) can be normalized, transforming the frequency f_{puls} into the dimensionless characteristic Strouhal number (Fig. 4) defined in Eq. 4:

$$Str = f_{puls} \cdot \frac{d_{eq}}{\bar{u}_{b,x}} \tag{4}$$

The equivalent diameter d_{eq} is obtained from the free flow area, perpendicular to the main flow direction, and $\bar{u}_{b,x}$ represents the mean volumetric velocity in axial direction at the burner exit. Their calculated values are given in Table 1.

For the investigated swirl burner types, the limit for the formation of coherent vortex structures is given by nearly constant combinations of Pu and Str. These critical parameter combinations Pu_{crit} and Str, found in the isothermal flow experiments, characterize the fluid dynamical conditions for the formation of a first-time ring-vortex structure in the burner near region. The hyperbolically decline can be derived from the idea that the ring-vortex formation is related to flow acceleration (du/dt) and the inertia of the flow ($\sim\bar{u}^2$). The maximum acceleration $(du/dt)_{max}$ of a sinousoidally in time forced velocity fluctuation is given by: $\hat{u} \cdot \omega_{puls} = \sqrt{2} \cdot u_{rms} \cdot 2\pi f_{puls}$. The product of the dimensionless parameters Pu_{crit} and Str comprises both effects on vortex formation, the flow acceleration and the inertia of the flow:

$$Pu_{crit} \cdot Str = \frac{u_{rms} \cdot f_{puls} \cdot d_{eq}}{\bar{u}_{b,x}^2}$$

With the help of this correlation the hyperbolically decline of Pu_{crit} with increasing Str ($Pu_{crit} \sim 1/Str$) can be understood. In previous work [10], this critical parameter combination for vortex formation ($Pu_{crit} \cdot Str$ = const, with x_{rw}/d_b = const) was obtained, too.

These results are very interesting for technical combustor applications, particularly with regard to periodic disturbances of the mass flow rate created by the burner construction itself (isothermal self-excited flow instabilities, e.g., vortex shedding) [17]. If the level of disturbance at the frequency given by the burner geometry and the flow rate is large enough ($>Pu_{crit}$), periodic coherent vortex rings close to the burner exit can be generated. A certain part of the mass flow rate through the burner, under premixed combustion conditions fuel gas/air mixture, will be rolled up in the vortex structure together with ambient medium. The reaction of the fuel in the vortex produces a fluctuation of the heat release and the flue gas production

TABLE 1

Data of the Investigations of Pulsated Isothermal Swirl Flows

$S_{0,th}$ [/]	δ [degree]	$\bar{V}_n$ [m³/h]	d_{eq} [mm]	$\bar{u}_{eff}$ [m/s]	$\bar{u}_{b,x}$ [m/s]	Re($\bar{u}_{eff}$, d_{eq}) [/]
0.21	15	45.0	40.88	10.64	10.27	25187
0.21	15	90.0	40.88	21.76	21.02	46169
0.45	30	45.0	38.41	12.05	10.43	28387
0.45	30	90.0	38.41	24.66	21.35	54864
0.79	45	45.0	34.43	14.99	10.60	31665
0.79	45	90.0	34.43	14.99	10.60	31665
0.79	45	90.0	34.43	30.68	21.69	61199

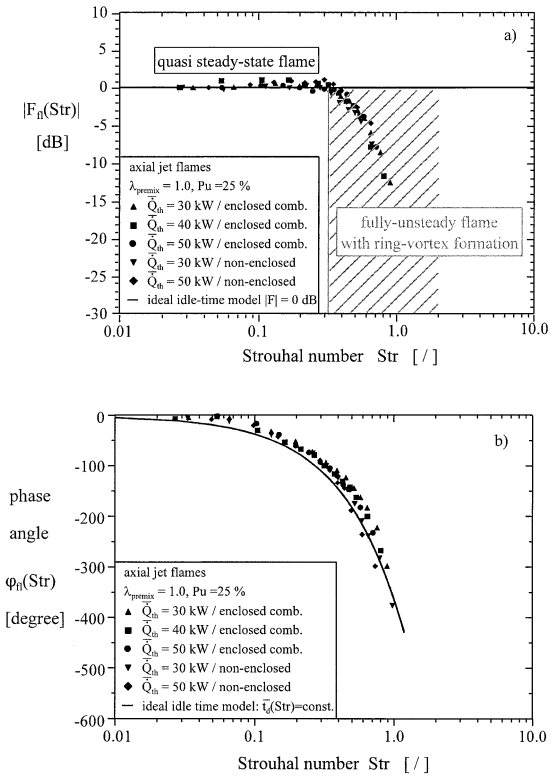

Fig. 5. Amplitude response $|F_{fl}(Str)|$ (a) and phase angle $\varphi_{fl}(Str)$ (b) of premixed axial jet flames.

rate, which could be seen as the starter of combustion driven oscillations (Fig. 1).

Investigations of the Flame Dynamics (Flame Transfer Function)

Flame Transfer Functions of Axial Jet Flames

For a better understanding of the flame response of premixed swirl flames, the flame characteristics of pulsated premixed axial jet flames (no swirl) are discussed first. In Fig. 5 the amplitude response (Fig. 5a) and the phase angle function (Fig. 5b) of turbulent, premixed jet flames are shown. These results are taken from previous investigations [4, 5]. Introducing a characteristic Strouhal number [4, 5] the dynamic jet flame characteristics could be described independent of mean thermal load $\bar{Q}_{th}(\sim Re)$ and encosure conditions. The amplitude responses show that the dynamic jet flame characteristics can be divided into two significantly different regions. Up to a certain Strouhal number (Str = 0.33 [4, 5]) the normalized amplitude response $|F|$ remains zero. This means that the pulsated flame behaves in every moment within the period of excitation as the steady-state flame with the corresponding mean thermal load $\bar{Q}_{th}$ and air equivalence ratio λ_{premix}, with regard to the OH*-radiation intensity of the entire flame. Therefore, this region of Strouhal number (Str < 0.33) is called "quasi steady-state flame" behavior, with regard to the mixing field and the reaction.

With increasing Strouhal number (Str > 0.33) the amplitude response decreases because of the formation and reaction of coherent ring-vortex structures [4, 5]. During their formation the ring-vortices strongly mix in additional ambient medium. Therefore, the fuel gas rolled up in the vortex is stronger diluted compared with the fuel gas concentration in the corresponding steady-state flame. In the non-enclosed case the dilution of the fuel gas share in the vortex with additional ambient air causes an increase of the local air equivalence ratio λ. Because of the strong dependence of the OH*-radiation intensity on the air equivalence ratio or flame temperature, as shown in Fig. 3a, this dilution leads to a decrease of the measured photomultiplier voltage $U_{pm}(t)$ during the vortex combustion in comparison to the corresponding steady-state flame. Therewith, the rms-value of the flame response $U_{pm,rms}(f_{puls})$ and the amplitude response $|F(f_{puls})|$ (see Eq. 2) decrease [4, 5].

In Fig. 5a and b the flame transfer functions are compared with the theoretical model of an ideal idle-time model, well known in control theory [13]. This simple model is characterized by the absence of any damping or amplification in the amplitude response $|F|$ (here: $U_{pm,rms}(f_{ref}) = U_{pm,rms}(f_{puls})$; $u_{b,rms}(f_{puls}) = \text{const.}$) and by a constant time delay for the phase angle ($\bar{t}_d(f_{puls}) = \text{const.}$). Up to the Strouhal number of about 0.33, characterized by quasi steady-state flame behavior, a good agreement can be observed between the measured amplitude response and the model. The measured phase angles show a smooth frequency dependence. Their course is in a good agreement with the theoretical model even for higher Strouhal numbers. This agreement demonstrates that the frequency-dependent phase angle of a jet flame

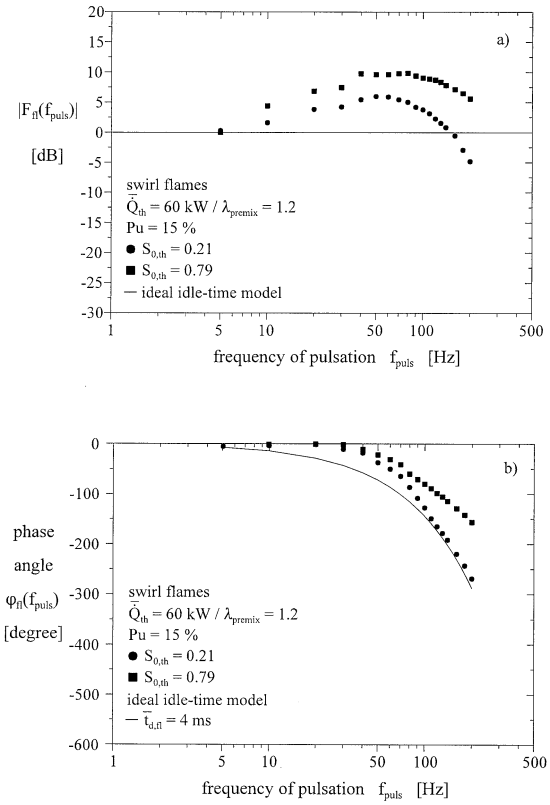

Fig. 6. Amplitude response $|F_{fl}(f_{puls})|$ (a) and phase angle $\varphi_{fl}(f_{puls})$ (b) of swirl flames with different swirl intensities $S_{0,th}$.

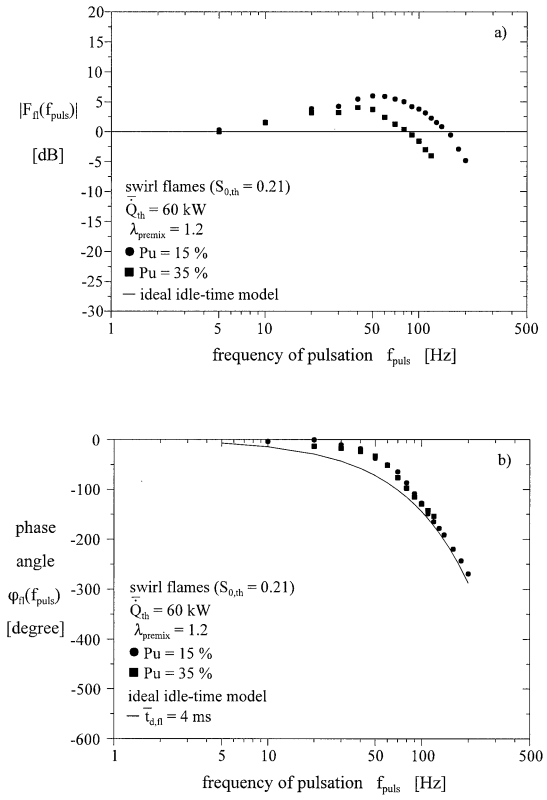

Fig. 7. Amplitude response $|F_{fl}(f_{puls})|$ (a) and phase angle $\varphi_{fl}(f_{puls})$ (b) of swirl flames with different pulsation levels Pu.

can be calculated with a corresponding, frequency-independent time delay. This time delay of a jet flame can be derived by the mean convective time delay for the transport of the mixture ($\bar{u}_b$) from the burner nozzle to the main reaction zone of the flame ($x_{OH,max}$) [4, 5]. With the help of this time delay of a given steady-state jet flame the dimensionless characteristic Strouhal number, used in Figs. 5a and b, was defined: $Str = f_{puls} \cdot x_{OH,max}/\bar{u}_b$) [4, 5].

Flame Transfer Functions of Swirl Flames

The dynamic flame characteristics of pulsated non-enclosed, premixed swirl flames with different swirl intensities and different pulsation levels are shown in Fig. 6 and Fig. 7, respectively. The flame transfer functions of the swirl flames are again compared with the theoretical model of an ideal idle-time model, described above.

The measured amplitude responses (Figs. 6a and 7a) display significant differences from this ideal idle-time model and therefore, from the dynamic jet flame characteristics. Up to a particular frequency the normalized fluctuations of the OH*-radiation intensities increase in comparison to the normalized fluctuations of the mass flow rate (see Eq. 2). With further increase of the frequency the measured amplitude frequency response decreases, so that the amplitude response of the flame transfer function for all measured premixed swirl flames is characterized by a maximum ($|F_{fl}(f_{puls})| > 0$ dB) at a particular frequency depending on two contrary effects, interpreted as follows:

1. The *decrease of the measured amplitude response*, which appears at higher frequencies, is caused by the formation and reaction of large-scale, coherent vortex structures. This effect was also observed at pulsated premixed axial jet flames [4, 5]. The detailed explanation for the decline in the amplitude response caused by the ring-vortex formation

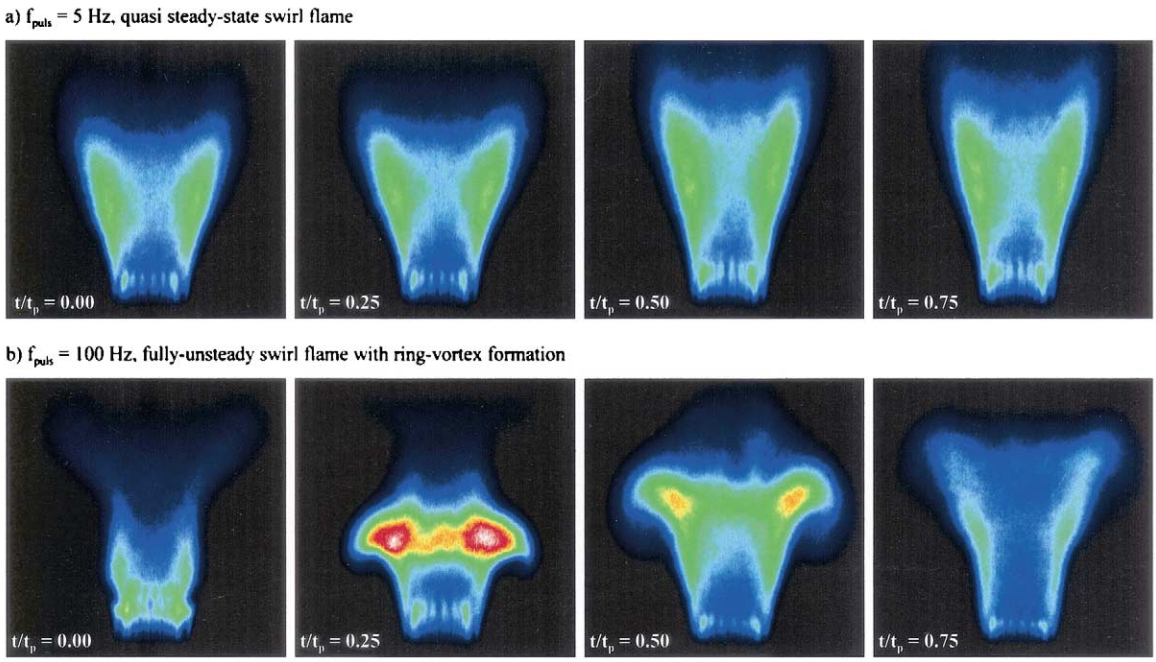

Fig. 8. Phase-correlated image-recording of a premixed swirl flame (swirl intensity: $S_{0,th} = 0.21$; mean thermal load: $\overline{\dot{Q}}_{th} = 60$ kW; air equivalence ratio: $\lambda_{premix} = 1.2$; pulsation level Pu = 15%; exposure time ICCD-Camera: $\Delta t_{exp} = 100\ \mu s$; 100 accumulations): (a) $f_{puls} = 5$ Hz, quasi steady-state swirl flame, (b) $f_{puls} = 100$ Hz, fully unsteady swirl flame with ring-vortex formation.

and reaction is already given in the previous section.

The existence of periodically formed and reacting toroidal vortices was also proved by phase-correlated image recording of the OH*radiation distribution in the flame as shown in Fig. 8b. Figure 8a displays different moments within a period of pulsation of a quasi steady-state swirl flame ($S_{0,th} = 0.21$; $\overline{\dot{Q}}_{th} = 60$ kW; $\lambda_{premix} = 1.2$), modulated at a frequency of $f_{puls} = f_{ref} = 5$ Hz and with a pulsation level of 15%. In every moment the flame geometry corresponds to a steady-state flame geometry with corresponding mean thermal load and air equivalence ratio. Increasing of the pulsation frequency f_{puls}, while keeping all the other operation parameters constant, the formation of large-scale vortices occurs (see Fig. 4). The formation of the toroidal vortices combined with a noticeable influence of the combustion process is illustrated in Fig. 8b for a frequency of pulsation of $f_{puls} = 100$ Hz.

The pulsation level Pu causes a significant influence on the frequency, at which the decrease of the amplitude response is detectable (Fig. 7a). With higher Pu values the formation of coherent vortex structures starts at lower frequencies of pulsation. This was found and discussed in the isothermal pulsated flow investigations (see Fig. 4). Therewith, the decline of the amplitude response starts at lower frequencies. The critical parameter combinations $(Pu_{crit}, f_{puls})_{flame}$ (or with normalized frequency $(Pu_{crit}, Str)_{flame}$), obtained by the flame transfer measurements, characterize—in contrast to the critical parameters found in the isothermal swirl flow experiments (see Fig. 4)—the beginning interaction of the fluid dynamically controlled vortex formation with the combustion process.

2. The *increase of the amplitude response* found for premixed swirl flames at moderate pulsation frequencies could not be detected for jet flames [4, 5]. Therefore, this difference in the flame behavior can be explained by the obvious differences of the entrainment characteristics between turbulent swirl flows and turbulent axial jet flows. It is well known that

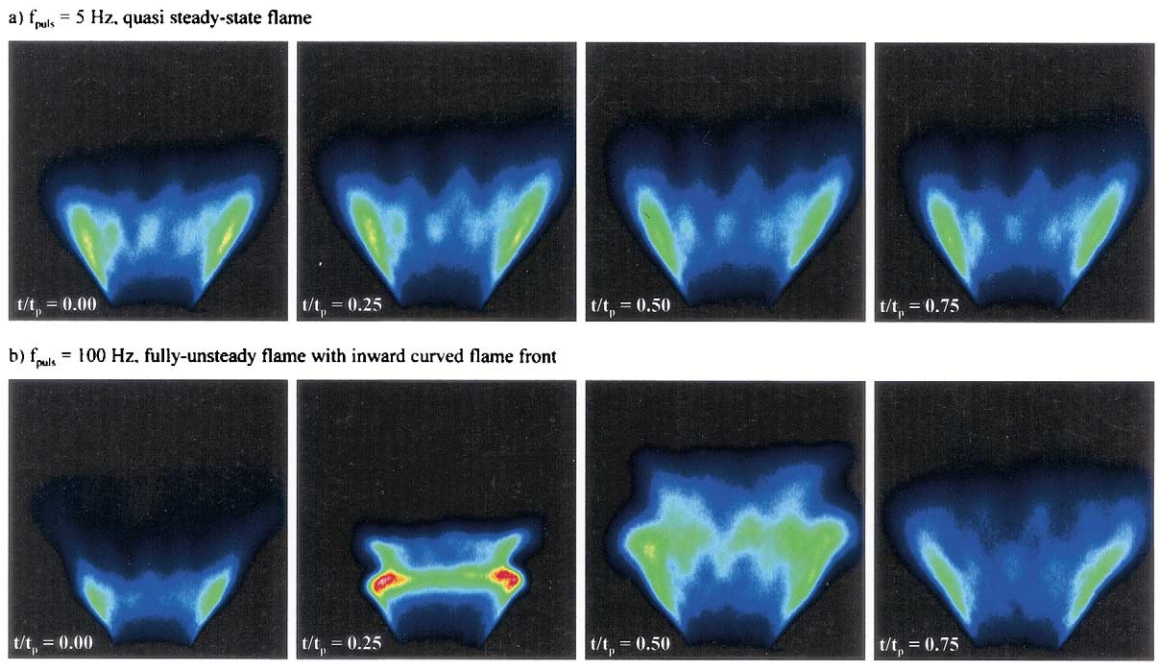

Fig. 9. Phase-correlated image-recording of a premixed swirl flame (swirl intensity: $S_{0,th} = 0.79$; mean thermal load: $\overline{Q}_{th} = 60$ kW; air equivalence ratio: $\lambda_{premix} = 1.0$; pulsation level Pu = 15%; exposure time ICCD-Camera: $\Delta t_{exp} = 100\ \mu s$; 100 accumulations): (a) $f_{puls} = 5$ Hz, quasi steady-state swirl flame, (b) $f_{puls} = 50$ Hz, fully unsteady swirl flame with inward curved flame front.

steady-state swirl flames entrain significantly more ambient medium than jet flames [14]. This additional entrainment of ambient air leads to an increase of the local air equivalence ratio of the combustion process combined with a decrease of the local combustion temperature. Therefore, the OH*-radiation intensity decreases with increasing swirl intensity of non-enclosed swirl flames (see Fig. 3b). An effect, that inhibits the additional entrainment of swirl flames during certain moments within the period of excitation, would cause the reaction of a particular amount of fuel gas at lower local air equivalence ratio than in the corresponding steady-state flame. This causes an increase of the fluctuation of the OH*-radiation intensities compared with the quasi steady-state flame ($U_{pm,rms}(f_{puls}) > U_{pm,rms}(f_{ref})$) and therewith, the amplitude response increases (see Eq. 2).

Furthermore, this interpretation model is underlined by the results shown in Fig. 6a: Increasing swirl intensity is correlated with an increase of entrainment of ambient air in the steady-state swirl flame [14] and so, as discussed above, with a decrease of the OH*-radiation intensities (see Fig. 3b). A restriction of entrainment in the pulsated case can result in a further increase of the OH*-radiation intensities related to the steady-state OH*-emissions with increasing swirl intensity. This is in good agreement with the higher values of the amplitude response measured at the pulsated swirl flames with higher swirl intensity.

An additional indication of the reaction of a part of the fuel gas at lower local air equivalence ratio than in the corresponding steady-state flames was found, analyzing the phase-correlated image recording of the pulsated swirl flame ($S_{0,th} = 0.79$, $f_{puls} = 50$ Hz), as shown in Fig. 9b. Figure 9a displays again the corresponding quasi steady-state swirl flame at a frequency of $f_{ref} = 5$ Hz. A moderate increase of the pulsation frequency up to $f_{puls} = 50$ Hz, while keeping all the other operation parameters constant, causes

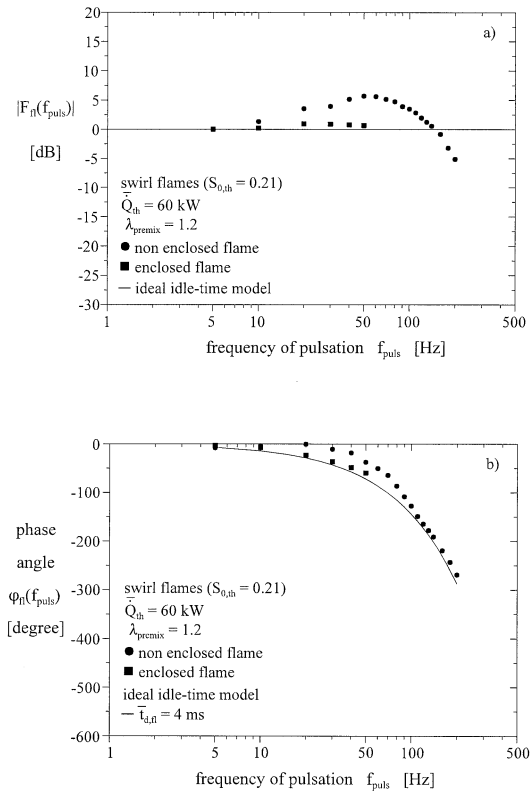

Fig. 10. Amplitude response $|F_{fl}(f_{puls})|$ (a) and phase angle $\varphi_{fl}(f_{puls})$ (b) of enclosed and non-enclosed swirl flames.

an increase of the amplitude response (see Fig. 6a). Parallel to this increase of the amplitude response the phase-correlated images show at particular moments within the period of excitation a thickened flame front, inward curved (Fig. 9b, $t/t_p = 0.25$). This means, that the reaction of a part of the fuel gas takes place in the region of the inner recirculation zone. This inward curved part of the fuel gas/air mixture, reacting in a region of hot temperature, could not observed in the quasi steady-state case (Fig. 9a). This reaction at higher temperature and possibly lower air equivalence ratio as in the quasi steady-state flame can therewith explain an increase of the amplitude response.

To strengthen the importance of this effect, described with the help of phase-correlated images, we measured the flame transfer function of an enclosed pulsated swirl flame. The results are shown in Fig. 10. The amplitude response of the enclosed case shows a nearly quasi steady-state flame characteristic in the moderate frequency range ($f_{puls} \leq 50$ Hz). This significant difference between the amplitude responses of non-enclosed and enclosed pulsated swirl flames can be explained with the reaction of a fuel gas share in the region of the inner recirculation zone. In contrast to the non-enclosed case the reaction of the entire fuel gas of the enclosed pulsated swirl flame takes place in regions with hot temperature (inner and outer recirculation zone) at any frequency of pulsation. Therefore, the reaction of the particular amount of fuel gas in the inner recirculation zone ($f_{puls} = 50$ Hz) does not amplify the fluctuation of the OH-radiation intensity in comparison to the corresponding enclosed quasi steady-state flame ($f_{ref} = 5$ Hz). The measured small increase of the amplitude response of the enclosed pulsated swirl flame could be caused by the fact, that the combustion chamber was not perfectly insulated (non-adiabatic). Therefore, temperature of the outer recirculation zone has to be lower than the temperature of the inner recirculation zone.

The effect of pulsation on the entrainment characteristics of swirl flows has been proved independently in previous investigations by isothermal time-resolved concentration measurements [18]. Within that work periodical fluctuations of the tracergas concentration in the inner recirculation zone of pulsated swirl flows, oscillating with the frequency of pulsation, were detected, while the ratio of the tracergas/air mixture mass flow through the burner was kept constant. For several moments within the period of excitation the tracergas concentration in the inner recirculation zone is higher than in the corresponding steady-state swirl flow. These results, discussed in detail in this previous paper [18] fit very well to the explanation of the effect, that increases the amplitude response of pulsated swirl flames as described above.

The superposition of these two effects, described in the two sections above, explains the frequency-dependent characteristics of the investigated pulsated, premixed swirl flames marked by a maximum value of the amplitude

response at a certain frequency f_{max} ($|F_{fl}(f_{max})| = |F_{fl}(f_{puls})|_{max}$). At this frequency the influence of periodically formed toroidal vortices on the combustion process exceeds the contrary, entrainment-restricting effect ($\rightarrow f_{max} = f_{crit}$).

The *phase angle functions* $\varphi_{fl}(f_{puls})$ of pulsated premixed swirl flames, displayed in Fig. 6b and 7b, differ considerably from the phase angle of the ideal idle-time model. As mentioned before, this model describes very well the phase angle functions of pulsated premixed axial jet flames [4, 5] and is characterized by a frequency-independent time delay ($\bar{t}_{d,fl}(f_{puls})$ = const.). This means (see Eq. 3), that for pulsated premixed swirl flames the time delay between the fluctuation of the mass flow rate at the burner exit and the resulting fluctuation of the heat release rate of the flame is strongly frequency-dependent ($\bar{t}_{d,fl}(f_{puls}) \neq$ const.), mainly in the low frequency range ($f_{puls} < 50$ Hz). The deviation from the ideal idle-time model seems to be caused by the same effect that increases the amplitude response of non-enclosed premixed swirl flames. This idea is underlined by the results shown in Fig. 10. In the case of the enclosed swirl flame the phase angle shows a very good agreement with the model for the low frequency range ($f_{puls} < 50$ Hz) in contrast to the non-enclosed swirl flame (Fig. 10b). In this range of moderate pulsation frequency the amplitude response of the non-enclosed swirl flame increases significantly, whereas the amplitude response of the enclosed swirl flame remains nearly constant (Fig. 10a).

For higher swirl intensities the phase angles decrease with a modest gradient and therewith, the swirl flame reacts on periodic excitation with a shorter time delay than in the case of lower swirl intensities (Fig. 6b). This dependency of the phase angle on swirl intensity can be explained by the increasing of the reaction density, resulting in a shorter flame length with higher swirl. Therefore, the mean convective time delay for the transport of the mixture from the burner nozzle to the main reaction zone of the flame is decreased with increasing swirl intensity. Assuming that the convective time delay dominates the total time delay occuring in this premixed flame system, the phase angle of a pulsated premixed flame decreases with increasing swirl intensity.

For different pulsation levels the phase angle functions $\varphi_{fl}(f_{puls})$ (Fig. 7b) show the same frequency-dependent behavior of premixed non-enclosed swirl flames ($\bar{t}_{d,fl}(f_{puls}) \neq$ const.). Though, there is no influence of the pulsation level on the time delay and phase angle, respectively. This result is very important for the physical understanding of the formation and amplification mechanism of self-excited combustion oscillations (see Fig. 1). The formation and reaction of an initial small ring-vortex coherent to the pressure fluctuation in the combustion chamber can cause the growing of the vortex and of the periodic part of the heat release rate and therewith, the pressure oscillations will be amplified. At the same time the phase angle between the oscillation of the mixture mass flow through the burner and the oscillation of the heat release rate of the flame (φ_{fl}) remains constant. This is essential for the in-phase interaction of the periodical disturbances inside the feedback cycle burner–flame-combustion chamber during the origin and amplification of self-sustained pulsating combustion. If the flame would change its phase angle φ_{fl} with increasing pulsation level Pu, the growth of combustion-driven pressure oscillations would inhibit itself.

CONCLUSIONS

The periodic formation and reaction of ring-vortex structures is a fundamental phenomenon combined with the appearance of self-excited flame-/pressure oscillations in technical combustion systems. Therefore, it is very important to get to know more details about the fluid dynamical conditions (f_{puls}, Pu, $\bar{m}_b$) of the ring-vortex formation in the isothermal flow field as well as under combustion conditions. Furthermore, a better understanding of the ring-vortex influence on the combustion process itself is necessary. In the present paper the fluid dynamical conditions for the formation of coherent ring-vortex structures in periodic modulated isothermal swirl flows are found to be determined by the combination of the dimensionless parameters Pu (pulsating level) and Str (normalized excitation frequency). The minimum excitation level for vortex formation decreases

with increasing Strouhal number Str (hyperbolical decline: $Pu_{crit} \sim 1/Str$).

Investigations of the frequency response of pulsated premixed swirl flames show two significant effects on their combustion process: The increase of the amplitude response (first effect) of pulsated swirl flames at moderate pulsation frequencies, that was not observed in pulsated premixed axial jet flames, can be explained by an effect, that inhibits the additional entrainment of ambient medium in comparison with the quasi steady-state premixed swirl flames at the same thermal load and air equivalence ratio. Therefore, the dynamic characteristics of non-enclosed premixed swirl flames show obvious differences to the jet flame behavior. The decreasing amplitude response with further increasing pulsation frequency represents the beginning influence of periodically formed toroidal vortices on the combustion process (second effect). Therefore, the amplitude responses of the investigated non-enclosed premixed swirl flames are characterized by maxima with positive values at a particular frequency.

In analogy to the results found by isothermal flow investigations, the increase of the disturbance level Pu causes the formation of ring-vortices at lower frequencies. Therewith, the interfering of coherent vortex structures on the combustion process starts at considerable lower frequencies. However, the critical frequencies for vortex formation in the isothermal swirl flows (Fig. 4) and for flame-vortex interaction in the flame measurements (Fig. 6a and 7a) at a constant pulsation level Pu are different. For the isothermal swirl flows the critical frequency and Strouhal number, respectively, characterizes the onset of vortex formation in the pulsated isothermal swirl flows close to the burner exit. In the flame investigations the critical frequency (maximum of the amplitude response) characterizes the beginning of a noticeable interaction of the fluid dynamically controlled vortex formation with the combustion process.

The obtained phase angles represent frequency-dependent time delays $\bar{t}_{d,fl}(f_{puls})$ of the investigated swirl flames on periodic excitation. This characteristic time delay of the flame considerably influences the in-phase interaction of periodical disturbances inside the feedback cycle "burner-flame-combustion chamber" (Fig. 1), which is necessary for coupling self-sustained pulsating combustion. Therefore, the knowledge of the phase angles of flames in dependence of $\bar{Q}_{th}$, λ_{premix}, $S_{0,th}$, and so forth is important for prediction models of the stability limits (feedback coupling) of premixed combustion systems.

The measured flame transfer functions of pulsated swirl flames can be used to check and to validate numerical simulations of the flame behavior for different excitation frequencies and amplitudes. Additionally, the determination of the flame dynamics is essential to improve prediction models for the formation of self-sustained combustion instabilities. Mention, that there are no other experimental data of periodically excited swirl flames in literature yet. Furthermore, the results obtained from the flame response measurements of pulsated flames (formation of large-scale vortices and their influence on combustion process and phase angles of the flame) enabled the development of measures to suppress the reactive ring-vortices and therewith, to control the formation of self-sustained combustion instabilities [19].

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